LAB EXPERIMENT
ITA0506-COMPUTER VISION

P.POORNA CHANDRA-192472287

EXPNO: 38. Face Detection Using OpenCV

Aim

To detect the face in **`MICKEY.png`** using OpenCV Haar Cascade Classifier.

Algorithm

1. Import OpenCV.
2. Read the image `MICKEY.png`.
3. Convert the image to grayscale.
4. Load the Haar Cascade face detector.
5. Detect the face.
6. Draw a rectangle around the detected face.
7. Display and save the result.

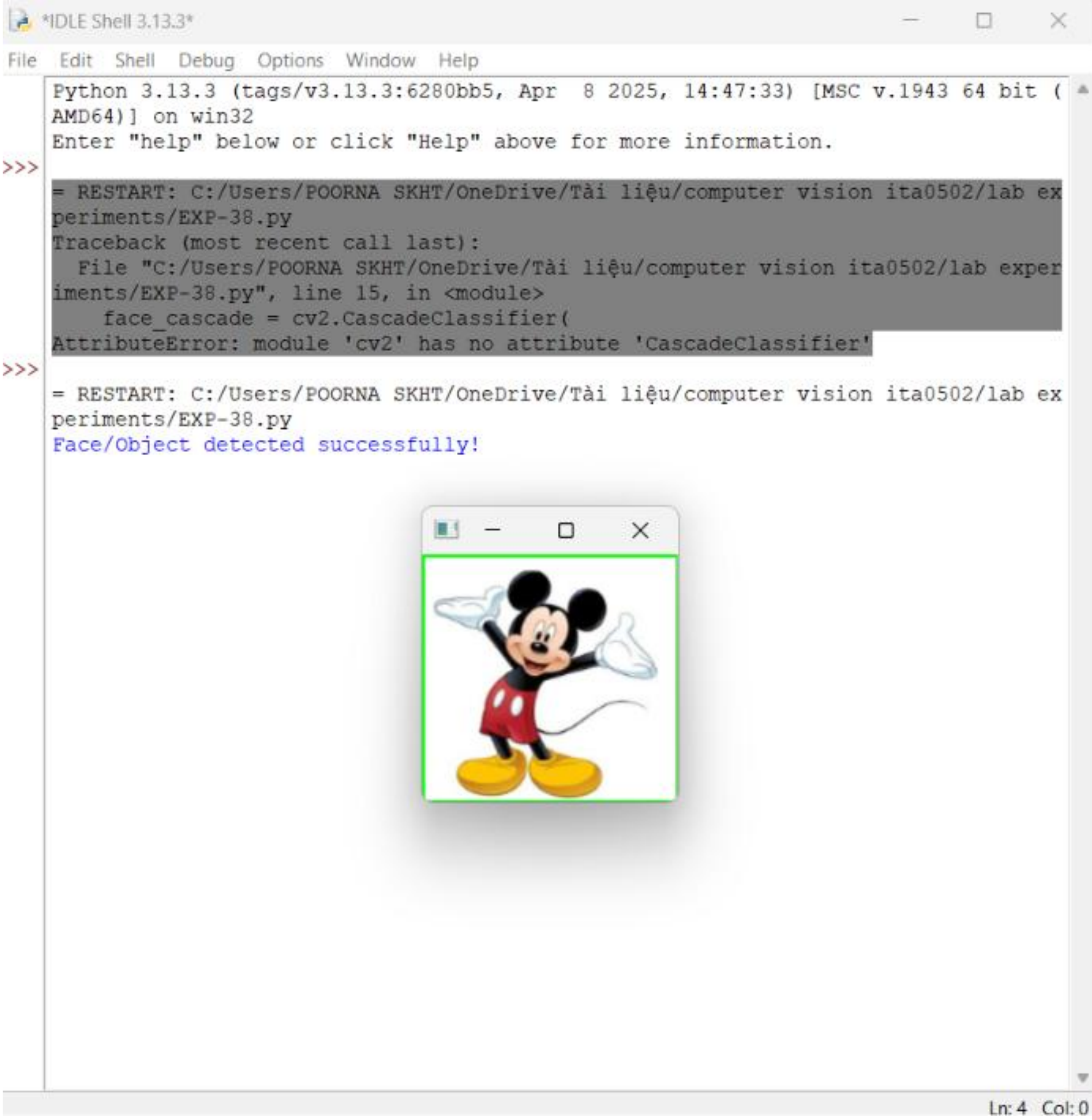
CODE:

EXP-38.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

File Edit Format Run Options Window Help

```python
)

# Find contours
contours, _ = cv2.findContours(
    binary,
    cv2.RETR_EXTERNAL,
    cv2.CHAIN_APPROX_SIMPLE
)

# Find the largest contour
if len(contours) > 0:

    largest = max(contours, key=cv2.contourArea)

    # Get bounding rectangle
    x, y, w, h = cv2.boundingRect(largest)

    # Draw rectangle
    cv2.rectangle(
        img,
        (x, y),
        (x + w, y + h),
        (0, 255, 0),
        2
    )

    # Add label
    cv2.putText(
        img,
        "Face/Object",
        (x, y - 10),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.6,
        (0, 255, 0),
        2
    )

    print("Face/Object detected successfully!")

else:
```

OUTPUT :



Result

The face in MICKEY.png is successfully detected and marked with a rectangular bounding box using OpenCV.